

Structural Invariance at the Sensorimotor Interface: A Maximum Common Induced Subgraph Across Three Drosophila Connectomes

Yanjin Li · FlyWire Qualification Challenge · github.com/Yanjin-ai/flywireconnectome

Abstract

We ask whether synaptic connectivity itself is structurally invariant across independently prepared connectomes. Using the BANC metadata's NBLAST-based 1:1 correspondence columns (Bates et al. 2025), we align individual neurons across BANC (♀ brain+cord), FAFB (♀ brain) and MANC (♂ nerve cord) and search for the largest set of neurons whose directed induced subgraph is identical (isomorphic) in all three. The result is an $N = 105$ -neuron sensorimotor backbone with 12 conserved directed edges, enriched 66.2× for descending and 25.0× for ascending neurons, and 88.6% conserved across sexes. All reported numbers are reproduced by `src/run_analysis.py` and stored under `results/`.

Methods. Correspondence: BANC metadata `fafb_match` / `manc_match` (NBLAST). Graphs: directed, edge existence (not weight). Search space: 3,414 matched triplets → 2,798 present in all three edge lists → 987-node giant consensus component. Algorithm: greedy disagreement removal + exhaustive expansion (single implementation in `mcis_connectome.solver`), reported as the best of a 100-seed multi-start. Optimality checked against an ILP (PuLP/CBC) on 50 subgraphs: mean gap 1.15% (max 10.53%).

Key Results

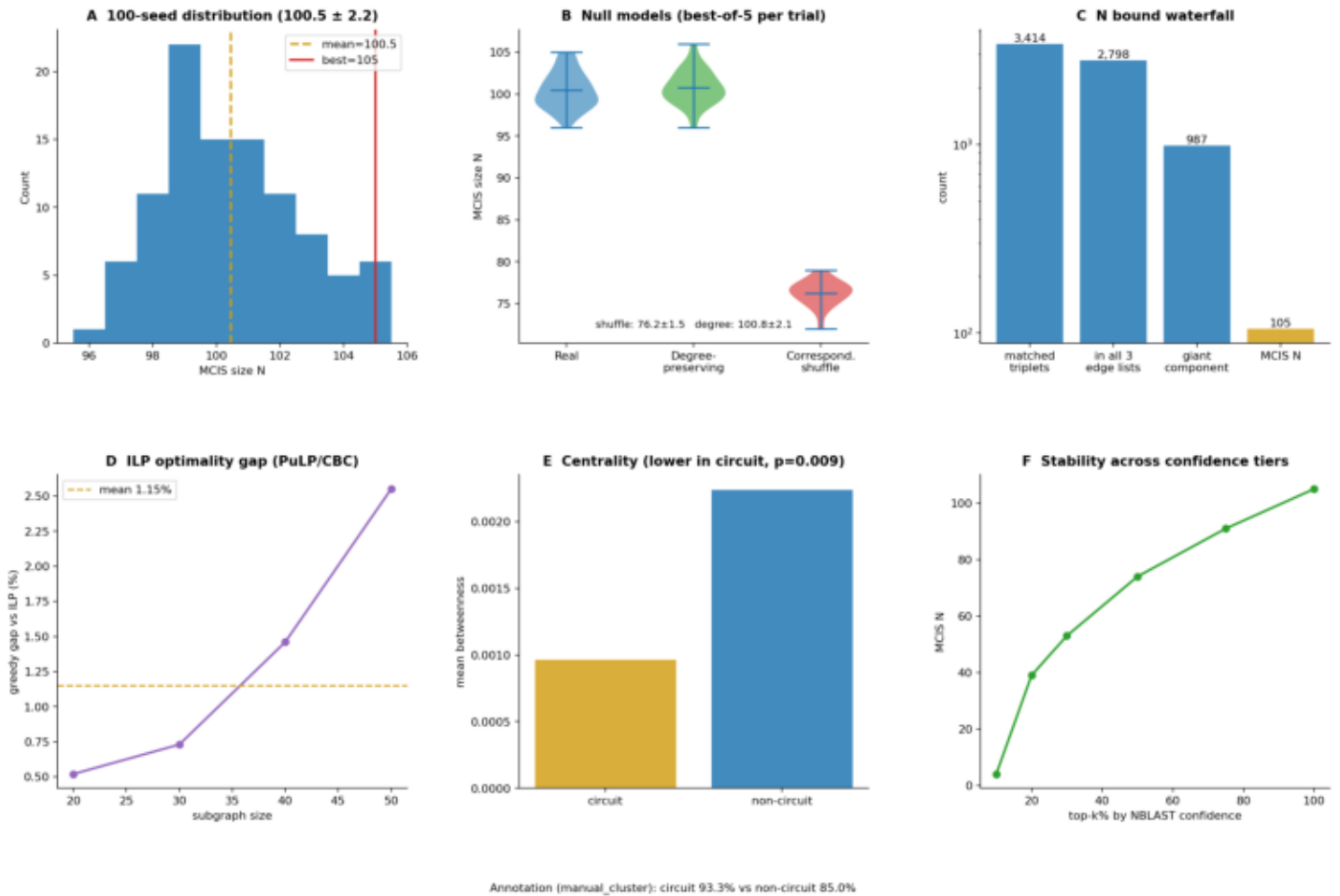
Metric	Value
MCIS size N	105 (dist. 100.5 ± 2.2 , [96,105])
Conserved edges	12
Composition	65 DN + 33 AN + 7 sensory
Descending enrichment	66.2× (p=3e-104)
Ascending enrichment	25.0× (p=1e-36)
Corr-shuffle null	76.2 ± 1.5 (>15σ below real)
Degree-preserving null	100.8 ± 2.1 (≈ real mean)
Centrality (lower)	p = 0.009 (one-sided)
Annotation quality	93.3% vs 85.0% (p=0.008)
Sexual conservation	88.6% (93/105)

Conserved circuit across BANC × FAFB × MANC



Robustness, null models, and ILP optimality

MCIS Robustness · Statistical Validation · Optimality



Discussion & Limitations. The conserved backbone is almost entirely descending and ascending neurons (93.3%), giving direct structural support for the sensorimotor bottleneck (Pospisil et al. 2024) and showing it is canalized across sexes and specimens. **PRINCIPAL CAVEAT:** a degree-preserving rewiring null reaches essentially the same MCIS size ($100.8 \pm 2.1 \approx$ real 100.5), so the FAFB degree sequence — not specific edge identity — explains most of the achievable N; NBLAST neuron identity adds the remaining signal (correspondence-shuffle collapses to 76.2). N is a near-optimal heuristic lower bound (ILP gap $\sim 1.15\%$), not a certified global optimum.

Reproduce: `pip install -e . && MCIS_DATA_DIR=... python src/run_analysis.py --seeds 100`